

YINGDAN SHI

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EDUCATION

Illinois Institute of Technology Ph.D. in Computer Engineering. Advisor: Prof. Ren Wang	Aug. 2024 - Jun. 2028 GPA: 4.0/4.0
Soochow University M.E. in Computer Technology. Advisor: Prof. Jingya Zhou	Sep. 2021 - Jun. 2024 GPA: 3.7/4.0
Hangzhou Normal University B.E. in Computer Science and Technology.	Sep. 2017 - Jun. 2021 GPA: 3.8/4.0

RESEARCH INTERESTS

Trustworthy post-training for LLMs and LLM-based agents, with a focus on removing, editing, and controlling learned knowledge and behaviors.

Keywords: Post-training, Safety Alignment, Agent Safety, Self-evolving Agents, Unlearning

RESEARCH EXPERIENCE

TIML Lab, Illinois Institute of Technology Aug. 2024 – Present
LLM Agents & Post-training

- **Trajectory Unlearning for LLM Agents** [[Under review, first author](#)]. Formulated *trajectory unlearning*, extending unlearning from factual knowledge to agent *behaviors*. Built agent unlearning benchmarks on ALFWorld and WebShop, and proposed **GiRPO**, a GRPO-based on-policy RL algorithm that reduces similarity to forgotten trajectories to **3%** on Qwen2.5-7B-Instruct while preserving task success rate.
- **Self-Unlearning for Self-Evolving Agents** [[Ongoing](#)]. Found that self-evolving agents silently re-acquire unlearned knowledge while evolving; designing self-unlearning inside the evolution loop to prevent re-memorization.
- **Knowledge Entanglement in LLMs** [[Ongoing](#)]. Modeled knowledge as hidden-state *subspaces* (not single vectors) and quantified inter-knowledge entanglement to guide targeted, low-collateral unlearning in LLMs.

Machine Unlearning & Trustworthy ML

- **Plasticity Collapse in Continual Unlearning** [[ECCV 2026, first author](#)]. Identified *plasticity collapse*: models gradually fail to forget new requests and re-memorize previously forgotten data. Traced it theoretically to accumulating geometric constraints in parameter space; validated on classification, DDPM, and LLaVA.
- **Nonlinear Unlearning via Mode Connectivity** [[ICML 2026, first author](#)]. Showed linear unlearning updates suffer from weight entanglement; proposed a nonlinear pathway along mode-connected curves, a plug-and-play add-on to existing methods that mitigates over- and under-forgetting.
- **Fake Forgetting** [[ICML 2026, first author](#)]. Used conformal prediction to show samples “forgotten” under accuracy metrics remain exposed; proposed an uncertainty-aware metric and unlearning framework to close the gap.
- **GNN Watermarking** [[ICML 2026, co-first author](#)]. Exposed ownership ambiguity and poisoning risks of backdoor watermarks; proposed an explanation-based watermark requiring no training-data manipulation.

SELECTED PUBLICATIONS

(* co-first author)

- [1] [Yingdan Shi](#), Xiang Xu, Kaize Ding, Alfred O. Hero and Ren Wang. “On the Plasticity Collapse in Continual Machine Unlearning.” **ECCV 2026**. [[Paper](#)][[Code](#)]
- [2] [Yingdan Shi](#) and Ren Wang. “Exploring Nonlinear Pathway in Parameter Space for Machine Unlearning.” **ICML 2026**. [[Paper](#)][[Code](#)]
- [3] [Yingdan Shi](#), Sijia Liu, Kaize Ding and Ren Wang. “Tackling Fake Forgetting through Uncertainty Quantification.” **ICML 2026**. [[Paper](#)][[Code](#)]
- [4] Jane Downer*, [Yingdan Shi*](#), Ziyang Liu*, Ren Wang and Binghui Wang. “Watermarking Graph Neural Networks via Explanations for Ownership Protection.” **ICML 2026**. [[Paper](#)][[Code](#)]
- [5] [Yingdan Shi](#) and Ren Wang. “Trajectory Unlearning on LLM-based Agents.” Under review, 2026. [[Paper](#)][[Code](#)]
- [6] Hyoseo Kim, [Yingdan Shi](#) and Ren Wang. “Unmasking Intent Blindness in Generative Engine Optimization.” Under review, 2026.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash

ML / LLM Frameworks: PyTorch, Hugging Face (Transformers, PEFT, TRL), verl, vLLM, DeepSpeed, PyTorch Geometric

Post-training: SFT, LoRA, DPO, GRPO, on-policy distillation, agentic RL

Infrastructure: Linux, Slurm, Docker, Git, Weights & Biases